Razorpay Notification System: Architecture Breakdown

Multi-Channel Delivery, Dynamic Gateway Routing & Sub-3s OTP Prioritization

FINTECH INFRASTRUCTURE
NOTIFICATION PLATFORM

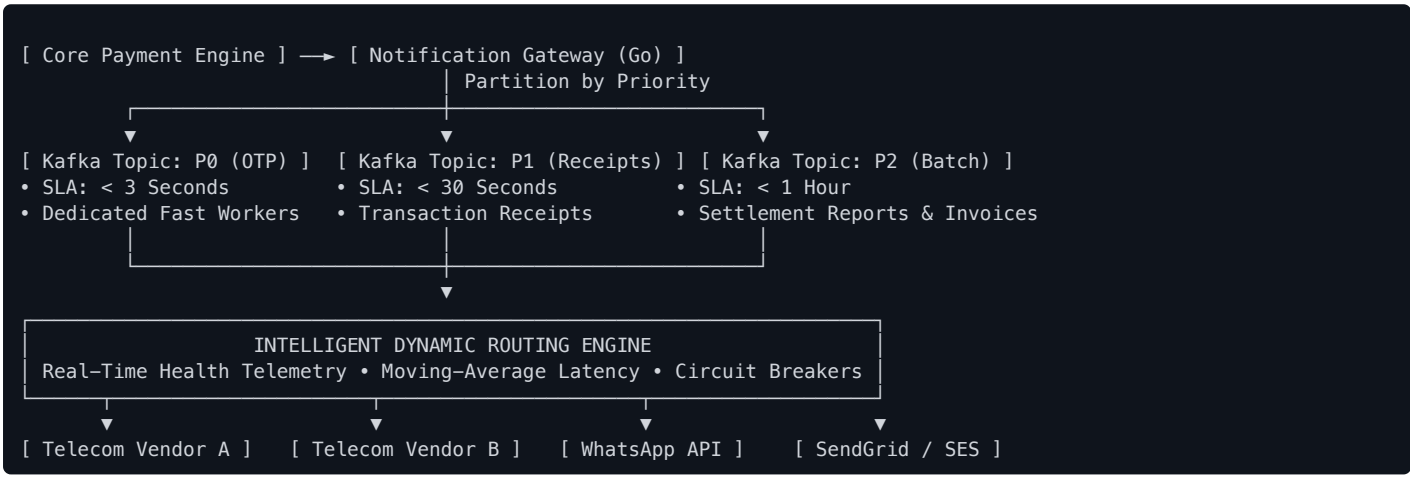
1. The Financial Impact of Delayed Notifications

At Razorpay, notifications are core transactional payment infrastructure. During bank checkout sessions, an authentication One-Time Password (OTP) must reach the user's mobile device within **under 3 seconds**. A 10-second delivery delay causes user anxiety, multiple OTP resend requests, and shopping cart abandonment.

The Operational Challenge

- **Multi-Channel Diversity:** Abstracting SMS, WhatsApp Business API, transactional email, webhooks, and mobile push into a unified API.
- **Strict SLAs:** Guaranteeing sub-3s delivery on critical OTPs while concurrently dispatching millions of asynchronous batch settlement reports.
- **Unreliable Telecom Carriers:** Operating across multiple telecommunication aggregators with volatile delivery rates and regional outages.

2. Platform Topology: Multi-Tier Queueing Architecture



3. Dynamic Carrier Routing & Automated Failover

Telecommunication SMS aggregators suffer regular transient fiber degradation and queue congestion. Razorpay continuously measures delivery quality in real time:

- **Rolling-Window DLR Health Checks:** Workers track Delivery Receipts (DLR) over a 60-second sliding window, evaluating success rates and delivery times per vendor.
- **Automated Circuit Breaker:** If Vendor A's delivery rate falls below 90% or p95 latency spikes past 4 seconds, the routing engine immediately trips a circuit breaker, shifting 100% of P0 OTP traffic to Vendor B in under 5 seconds.
- **Smart Channel Fallback:** If SMS delivery stalls due to cellular network congestion, the system automatically dispatches the alert over WhatsApp Business API.

4. Distributed Idempotency: Eliminating Double Alerts

Sending duplicate payment debit alerts causes severe panic. Razorpay enforces strict idempotency using distributed Redis locks:

Atomic Redis SETNX Lease
Key: `notif:{channel}:{idempotency_token}`.
Enforces an atomic 24-hour lease. Concurrent retries detect existing leases and short-circuit immediately without sending duplicate messages.

Rate Limiting & Anti-Bombing
Token-bucket limiters cap OTP dispatches per phone number (e.g. max 3 attempts per 10 minutes), neutralizing SMS bombing and brute-force verification attacks.

5. Notification Strategy Comparison Table

Architectural Vector	Standard SaaS Setup	Razorpay Fintech Engine
Queue Architecture	Single shared FIFO queue (RabbitMQ)	**Strict Priority Kafka Topics (P0, P1, P2)**
Vendor Redundancy	Single telecom vendor (Single SPoF)	**Multi-vendor active-active with live failover**
OTP Latency SLA	Best effort (15s – 60s)	**Sub-3 seconds guaranteed (< 1% drop)**
Cross-Channel Fallback	None (Fails on SMS drop)	**Automatic SMS ——► WhatsApp failover**

6. High-Throughput Delivery Receipt (DLR) Ingestion

Telecommunication gateways stream millions of asynchronous DLR webhooks reporting tower delivery timestamps:

- **Stateless Edge Ingestion:** Lightweight Go reverse proxies receive carrier webhooks, write payloads directly to partitioned Kafka topics, and return 200 OK in under 5ms.
- **Asynchronous State Machine:** Dedicated reconciliation workers update delivery status in relational audit databases, computing real-time vendor latency metrics and merchant billing analytics.

7. Key Architectural Takeaways for Fintech Notifications

Core Principles Distilled from Razorpay's Notification Platform

- **Strictly Segregate Priority Tiers:** Never mix time-sensitive authentication OTPs with bulk marketing campaigns. Dedicated queue partitions prevent fatal head-of-line blocking.
- **Dynamically Route Around Degraded Carriers:** Telecommunication vendor reliability is inherently erratic. Track real-time delivery receipt metrics and automate multi-vendor circuit breaker failovers.
- **Enforce Atomic Distributed Idempotency:** Duplicate financial notifications erode consumer trust. Always protect delivery pipelines with atomic Redis distributed locks.
- **Layer Cross-Channel Fallbacks:** When cellular SMS delivery lags during regional network congestion, immediately re-route messages across IP-based channels like WhatsApp Business.